

VIP: TOWARDS UNIVERSAL VISUAL REWARD AND REPRESENTATION VIA VALUE-IMPLICIT PRE-TRAINING

ICLR(International Conference on Learning Representations) 2023

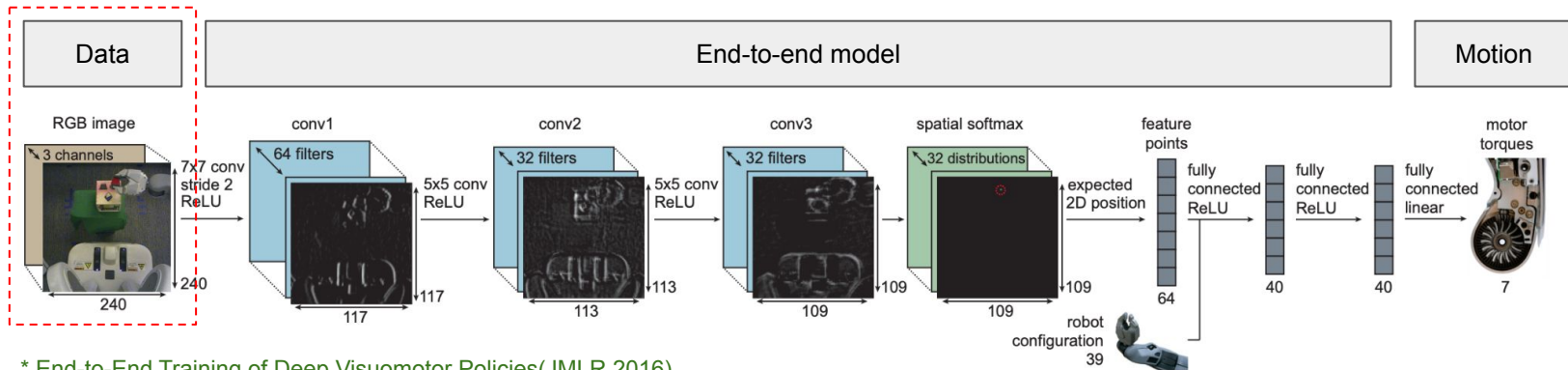
Introduction

Previous work

: How do we train a robot to complete a manipulation task?

- A standard and widely used approach is to train an end-to-end model from scratch using data from the **same domain**

→ However, this can be prohibitively data intensive and severely limits generalization



: A long-standing challenge in robot learning is to develop robots that can learn a diverse and expanding set of manipulation skills from sensory observations(e.g., vision).

→ Effective task representation and specification for downstream policy learning demands **scalable and generalizable representation learning and reward learning**

* R3M
(Robust Visual Encoder)

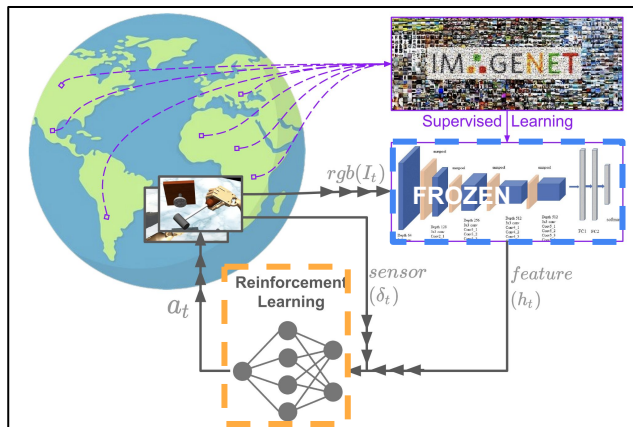
* VIP

Introduction

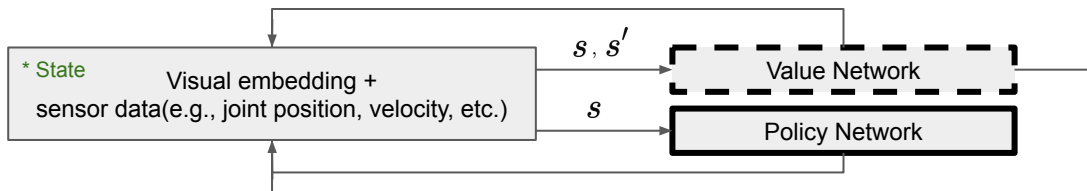
Previous work

: Well-shaped fixed reward function

- RRL: Resnet as representation for Reinforcement Learning [PMLR, 2021]



* A2C(Actor-Critic)



$$\eta(\pi) = \mathbb{E}_{\pi, \mathcal{M}} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right] \quad \text{where } r_t = \begin{cases} 1 & \text{if task success at time } t \\ 0 & \text{otherwise} \end{cases} \quad (\text{sparse reward setting})$$

where $P(s_{t+1} | s_t, a_t, s_{t-1}, a_{t-1}, \dots, s_0, a_0) = P(s_{t+1} | s_t, a_t)$

$$V^\pi(s) = \mathbb{E}_{\pi, \mathcal{M}} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s \right], \quad Q^\pi(s, a) = \mathbb{E}_{\mathcal{M}} [\mathcal{R}(s, a)] + \mathbb{E}_{s' \sim \mathcal{T}(s, a)} [V^\pi(s')]$$
$$A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$$

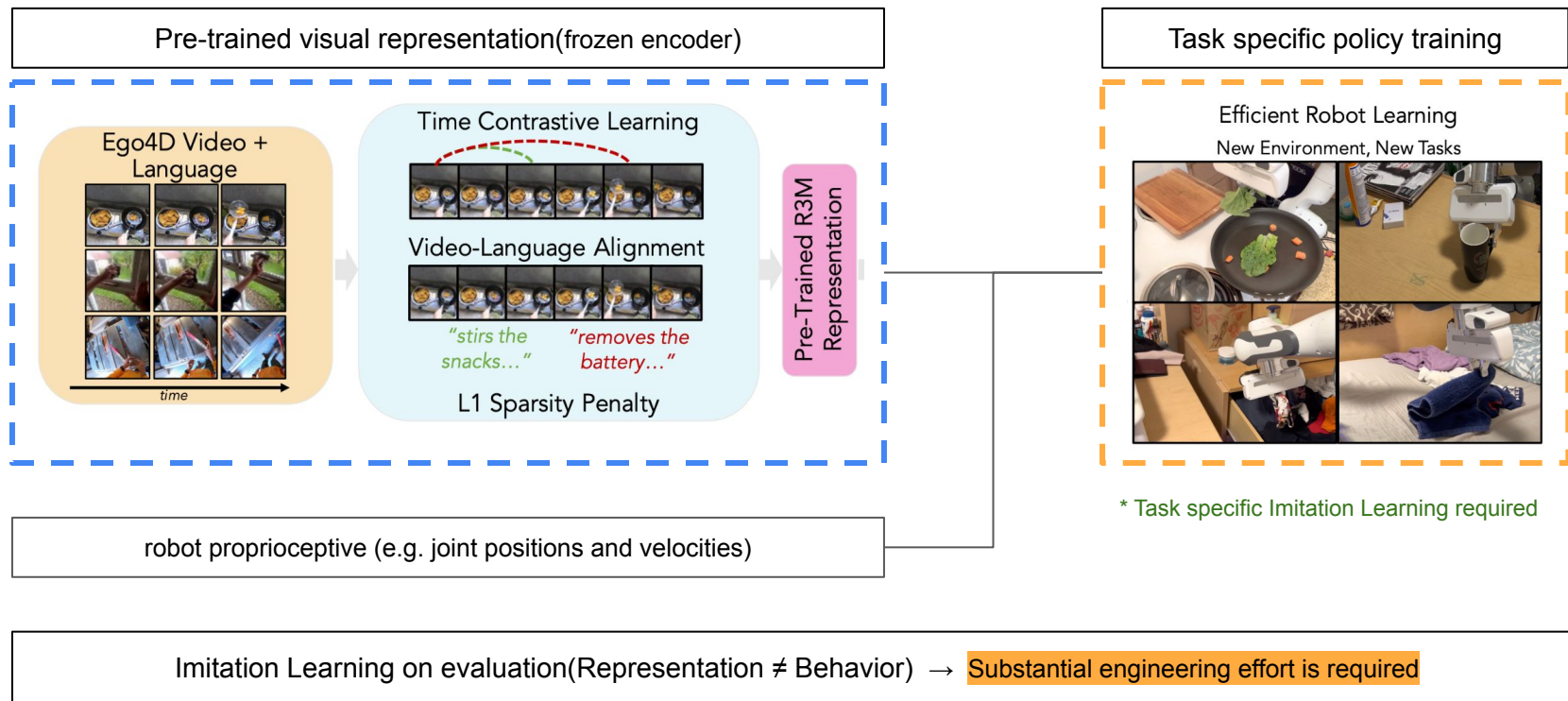
$$\nabla \eta(\pi) = g = \frac{1}{NT} \sum_{i=0}^N \sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t^i | s_t^i) \hat{A}^{\pi}(s_t^i, a_t^i, t), \quad \theta_{t+1} \approx \theta_t + \alpha [\nabla_{\theta} \log \pi_{\theta}(a | s) \delta_v]$$

Imitation Learning(Representation $\neq$ Behavior) $\rightarrow$ Substantial engineering effort is required

Introduction

Previous work

- : Visual imitation learning(IL) from demonstrations(Behavior Cloning)
- R3M: A Universal Visual Representation for Robot Manipulation [CoRL, 2022]

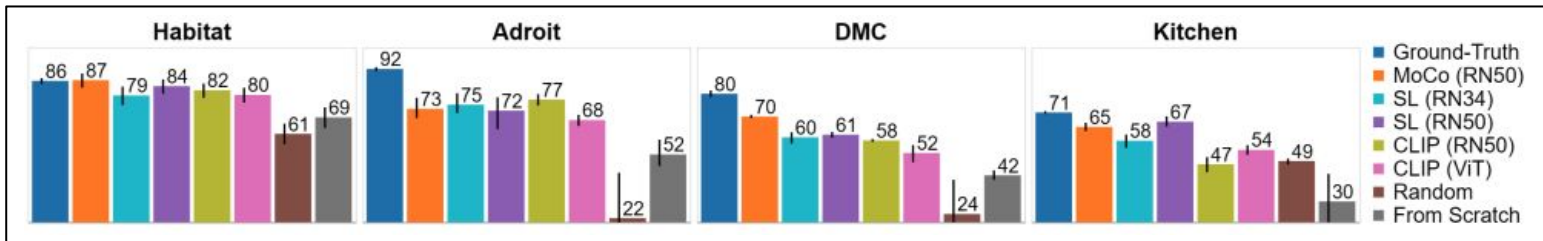
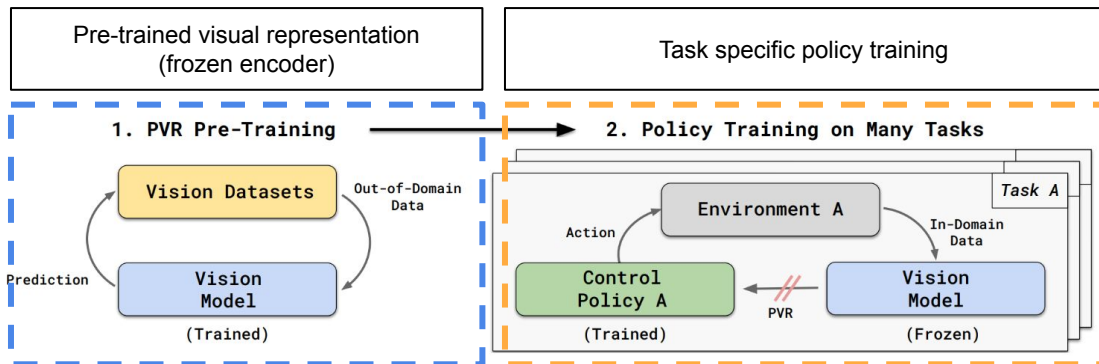
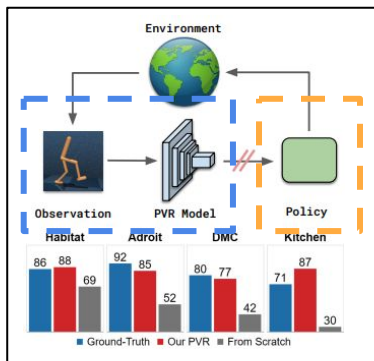


Introduction

Previous work

: Visual imitation learning(IL) from demonstrations

- The (Un)Surprising Effectiveness of Pre-Trained Vision Models for Control [PMLR, 2022]



Representation $\neq$ Behavior $\rightarrow$ Substantial engineering effort is required

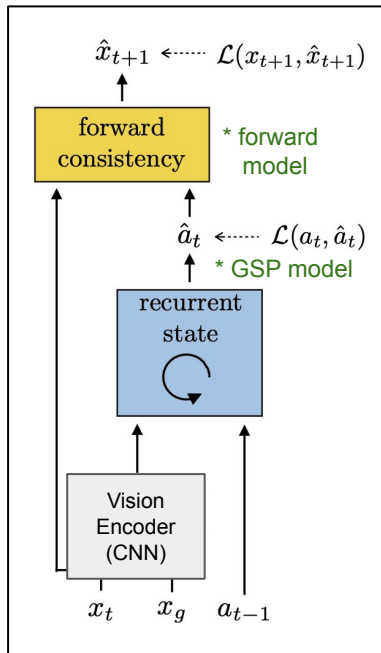
Introduction

Previous work

: Providing a goal image that captures the desired visual changes to the environment.

- ZERO-SHOT VISUAL IMITATION [ICLR, 2018]

* Self-supervised goal-conditioned skill policy



Dataset	Sequence(Observation - Action) $\longrightarrow$ policy $a = \pi_E(s) \longrightarrow \vec{a}_\tau = \pi(x_i, x_g; \theta_\pi)$
Loss function	$\min_{\theta_\pi, \theta_f} \ x_{t+1} - \tilde{x}_{t+1}\ _2^2 + \lambda \ x_{t+1} - \hat{x}_{t+1}\ _2^2 + \mathcal{L}(a_t, \hat{a}_t) \quad \text{s.t.} \quad \begin{aligned} \tilde{x}_{t+1} &= f(x_t, a_t; \theta_f) \\ \hat{x}_{t+1} &= f(x_t, \hat{a}_t; \theta_f) \\ \hat{a}_t &= \pi(x_t, x_{t+1}; \theta_\pi) \end{aligned}$ ① Pre-train the forward model with only the first term ② Pre-train π (behavior cloning) ③ Fine-tune jointly : The forward model f might not be good in the beginning, and hence could make the gradient updates noisier for π $\mathcal{L}(a_t, \hat{a}_t) = p(a_t x_t, x_{t+1}) \log(\hat{a}_t)$ * Discrete action classification (Cross-entropy)



Initial Image



Step-7



Step-14



Step-21



Step-28



Step-35



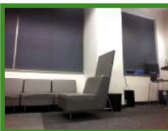
Step-42



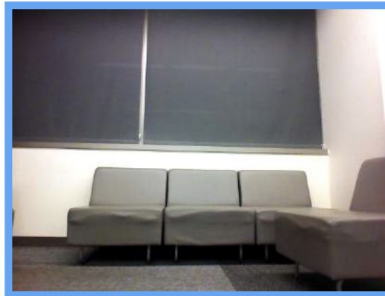
Step-44



Step-51



Final Image (Step-59)



Fixed Goal Image

Introduction

Previous work

RRL (Reinforcement Learning with ResNet-based Representation)

- : Generalized visual representations(supervised learning) + **Imitation learning** + Sparse binary success indicator(reward signal)
 - Stable policy learning can be achieved

R3M (Reusable Representations for Robotic Manipulation)

- : ResNet-Transformer encoder on the Ego4D dataset using self-supervised learning(SSL) + Imitation learning
 - Strong general-purpose representations

The (Un)Surprising Effectiveness of Pre-Trained Vision Models for Control

- : Supervised and self-supervised visual models + Control policy learning in simulated RL tasks
 - Strict **self-supervised** representations often **outperform supervised** ones in downstream control

Zero-Shot Visual Imitation

- : Goal-conditioned Skill Policy (GSP) training by using goal images alone (w/o access to expert actions or extrinsic rewards)
 - Achieves **true zero-shot imitation**(enables imitation at test time purely from visual observations)

→ Reward functions and policy still need to be learned separately.

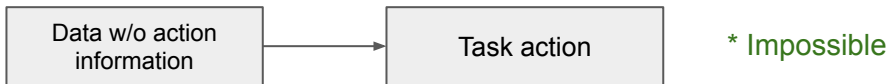
→ Pre-trained visual representations(embedding distance) do not produce effective reward functions

* Doesn't contains control-relevant features

Introduction

Contribution

: Problems



- Is it possible to

1. enabling zero-shot control without task-specific reward design or imitation learning
2. by deriving a *universal* visual reward function entirely from general visual representation of out-of-domain data?

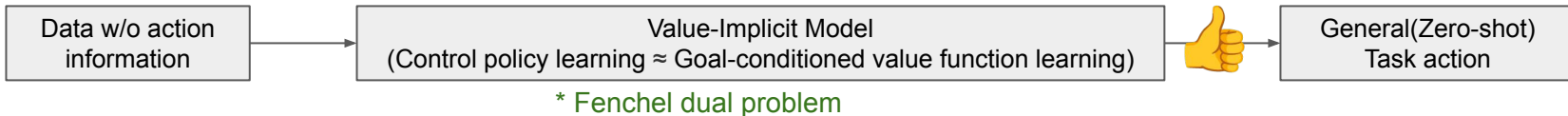
: Difference

- Our idea philosophically diverges from all prior works:

Instead of taking what worked the best for CV tasks and “hope for the best” in visual control,



We propose a more principled approach of using RL itself as a pre-training mechanism for RL



Related work

Out-of-Domain Representation Pre-Training for Control

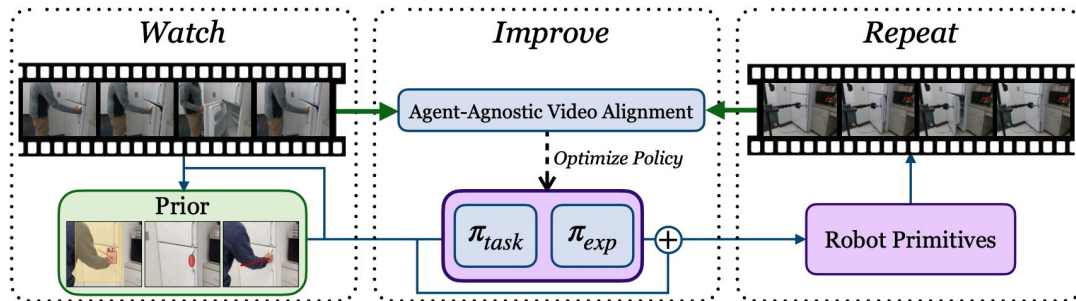
: R3M

- Pre-trained on the Ego4D video dataset (temporal information) by using time-contrastive learning.
- Additionally requires video textual descriptions to align its representation.

→ Prior works primarily **re-purpose existing objectives and models** and **do not address the reward specification challenge**.

Perceptual Reward Learning from Human Videos

: Human-to-Robot Imitation in the Wild [RSS, 2022]



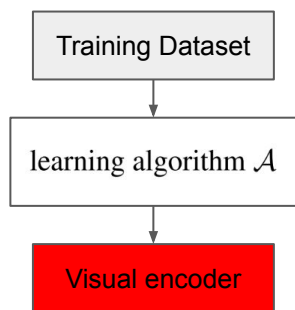
- First "watch" and obtain human priors(hand movement and object interactions) (Extracting human priors and convert to robot priors)
- Second, "repeat" trying to achieve task success and explore around the prior (Fix exploration-policy and update task-policy)
- Finally, "improve" our task policy (Leveraging reward from agent-agnostic video alignment model)

→ Require **task-specific** human demonstration videos paired with robot videos solving the same task

→ **Cannot leverage** the large amount of "in-the-wild" human videos readily available

Problem Setting and Backgrounds

Out-of-Domain Pre-Training Visual Representation



* Out- of-Domain video dataset (Image dataset where $h_i = 1$)

$$D = \{v_i := (o_1^i, \dots, o_{h_i}^i)\}_{i=1}^N, \text{ where each } o \in O := \mathbb{R}^{H \times W \times 3}$$

* Image data

$$\phi := \mathcal{A}(D) : \mathbb{R}^{H \times W \times 3} \rightarrow K, \text{ where } K \text{ is the embedding dimension}$$

Representation Evaluation

: Given a choice of representation ϕ , every evaluation task can be instantiated as a Markov decision process(reconfigured)

parameters of ϕ are frozen during policy learning

$$\mathcal{M}(\phi) := (\underbrace{\phi(O)}_{\text{state space}}, \underbrace{A}_{\text{action space}}, \underbrace{R(o_t, o_{t+1}; \phi, g)}_{\text{goal-conditioned value-function}}, \underbrace{T, \gamma, g}_{\substack{\text{discount factor} \\ \text{transition function}}}, \underbrace{\text{goal image}})$$

$$R(o_t, o_{t+1}; \phi, \{g\}) := \mathcal{S}_\phi(o_{t+1}; g) - \mathcal{S}_\phi(o_t; g) := (1 - \gamma)\mathcal{S}_\phi(o_{t+1}; g) + (\gamma\mathcal{S}_\phi(o_{t+1}; g) - \mathcal{S}_\phi(o_t; g))$$

$$\mathcal{S}_\phi(o_t; g) := -\|\phi(o_t) - \phi(g)\|_2$$

a distance function in the ϕ -representation space

→ we want to learn a policy $\pi : \mathbb{R}^K \rightarrow A$ that outputs an action based on the embedded observation $a \sim \pi(\phi(o))$.
there is only one goal to be attempted per task

Method

VALUE-IMPLICIT PRE-TRAINING

: FOUNDATION - SELF-SUPERVISED VALUE LEARNING FROM HUMAN VIDEOS

(Theoretical definition)

- Human videos are out-of-domain data for robots, but in-domain for learning a goal-conditioned policy π_H over human actions,

$$a^H \sim \pi^H(\phi(o) \mid \phi(g)), \text{ for some human action space } \mathcal{A}^H.$$

: Since human videos naturally contain goal-directed behavior, it is reasonable to solve an offline goal-conditioned RL problem over the space of human policies and then extract the learned visual representation.

* The distribution over observation-action pairs induced by a human policy conditioned on a goal g

* Unobserved 'true' human action taken to transition

$(o_h^i, \tilde{a}_h^i, o_{h+1}^i)$ in the dataset D

$$\max_{\pi_H, \phi} \mathbb{E}_{\pi_H} \left[\sum_t \gamma^t r(o; g) \right] - D_{\text{KL}}(d^{\pi_H}(o, a^H; g) \| d^D(o, \tilde{a}^H; g))$$

* Difference between two distribution

* The distribution over observation-dummy action pairs induced by a Dataset conditioned on a goal g

: While this objective is mathematically sound and encourages learning a conservative π_H , it is seemingly implausible because the offline dataset does not come with any action labels nor can $\mathcal{A}^H$ be concretely defined in practice.

→ Possible to derive dual objective over a value function that does not depend on any action label in the offline dataset.

Method

VALUE-IMPLICIT PRE-TRAINING

: FOUNDATION - SELF-SUPERVISED VALUE LEARNING FROM HUMAN VIDEOS

* where Lagrangian is convex-concave function

(Fenchel Duality) $\max_x f(x) \quad \text{s.t.} \quad g_i(x) \leq 0 \longrightarrow \mathcal{L}(x, \lambda) = f(x) - \sum_i \lambda_i g_i(x) \longrightarrow d(\lambda) = \max_x \mathcal{L}(x, \lambda) \longrightarrow \min_{\lambda \geq 0} d(\lambda)$

$$\max_x f(x) - D(x) \longrightarrow \min_V \max_x \text{some dual form}$$

(Policy space optimization) $\max_{\pi_H, \phi} \mathbb{E}_{\pi_H} \left[\sum_t \gamma^t r(o; g) \right] - D_{\text{KL}}(d^{\pi_H}(o, a^H; g) \| d^D(o, \tilde{a}^H; g))$

* Expected cumulative reward

* Divergence from data(prevent policy gets far from dataset)

(Dual value-function objective)

$$\max_{\phi} \min_V \mathbb{E}_{p(g)} [(1 - \gamma) \mathbb{E}_{\mu_0(o; g)} [V(\phi(o); \phi(g))] + \log \mathbb{E}_{(o, o'; g) \sim D} [\exp(r(o, g) + \gamma V(\phi(o'); \phi(g)) - V(\phi(o), \phi(g)))]]$$

1. Actions do not appear in the objective
2. All expectations can be sampled using the offline dataset

$r(o, g) = \mathbb{I}(o == g) - 1$ Set reward function as Indicator to extracts a general notion of goal-directed task progress

- Can be self-supervised with an appropriate choice of reward function
- Provides a constant negative reward when o is not the provided goal g, and does not require any task-specific engineering. $\longrightarrow$ Visual encoder and value function could be achieved!

Method

VALUE-IMPLICIT PRE-TRAINING

: ANALYSIS - IMPLICIT TIME CONTRASTIVE LEARNING

$$\max_{\phi} \min_V \mathbb{E}_{p(g)} [(1 - \gamma) \mathbb{E}_{\mu_0(o;g)} [V(\phi(o); \phi(g))] + \log \mathbb{E}_{(o,o';g) \sim D} [\exp(r(o,g) + \gamma V(\phi(o'); \phi(g)) - V(\phi(o), \phi(g)))]]$$

Above equation can be understood as a novel implicit temporal contrastive learning objective

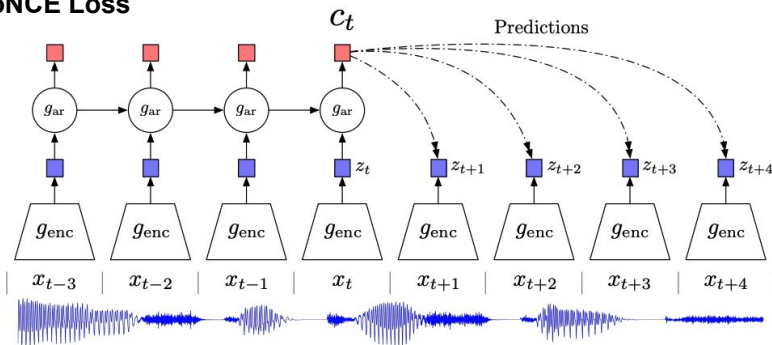
→ Acquires temporally smooth embedding distance over video sequences

assuming that the optimal V^* is found:

$$\min_{\phi} \mathbb{E}_{p(g)} \left[(1 - \gamma) \mathbb{E}_{\mu_0(o;g)} [-V^*(\phi(o); \phi(g))] + \log \mathbb{E}_{D(o,o';g)} \left[\exp \left(\tilde{\delta}_g(o) + \gamma V(\phi(o'); \phi(g)) - V(\phi(o), \phi(g)) \right) \right]^{-1} \right]$$

If we think of $V^*(\phi(o); \phi(g))$ as a similarity metric in the embedding space,
the expression above resembles the **InfoNCE** time contrastive learning objective

InfoNCE Loss



* Positive : observation at t+1

* Anchor : observations until t ($x_{t-n}, \dots, x_t$)

$$\mathcal{L}_N = -\mathbb{E}_X \left[\log \frac{f_k(x_{t+k}, c_t)}{\sum_{x_j \in X} f_k(x_j, c_t)} \right]$$

* Negative : Non-positive observations $x_j \neq x_{t+k}$

→ Find out future from current and past observations

Method

VALUE-IMPLICIT PRE-TRAINING

: ANALYSIS - IMPLICIT TIME CONTRASTIVE LEARNING

* Sampled goal $\rightarrow$ anchor

* Distribution of initial observations $\rightarrow$ positive

* Arbitrary transition samples $\rightarrow$ mostly negative

$$\min_{\phi} (1 - \gamma) \mathbb{E}_{p(g), \mu_0(o; g)} \left[-\log \frac{e^{V^*(\phi(o); \phi(g))}}{\mathbb{E}_{D(o, o'; g)} \left[\exp \left(\tilde{\delta}_g(o) + \gamma V^*(\phi(o'); \phi(g)) - V^*(\phi(o), \phi(g)) \right) \right]^{\frac{-1}{(1-\gamma)}}} \right]$$

In contrast to standard single-view time contrastive learning (TCN), in which the positive observations are temporally closer to the anchor observation than the negatives,

$\rightarrow$ Equation above has the **positives** to be as temporally far away as possible, **namely the initial frame** in the the same video sequence, and the **negatives to be middle frames** sampled in between.

(The lack of explicit repulsion) $\mathcal{L}_{\text{contrastive}} = -\log \frac{\exp(\text{sim}(x, x^+))}{\exp(\text{sim}(x, x^+)) + \sum_{x^-} \exp(\text{sim}(x, x^-))}$; non-existing term in Dual value-function objective

* Help 'negative' gets far from 'anchor'

(TD-Error) One-step temporal-difference error in the representation space

$$\tilde{\delta}_g(o) + \gamma V^*(\phi(o'); \phi(g)) - V^*(\phi(o), \phi(g))$$

While training, observations close to goal $\rightarrow$ smaller TD error
observations far from goal $\rightarrow$ bigger TD error

$\rightarrow$ Instead of an explicit constraint as in standard (time) contrastive learning, the repulsion of the negative observations is an implicit

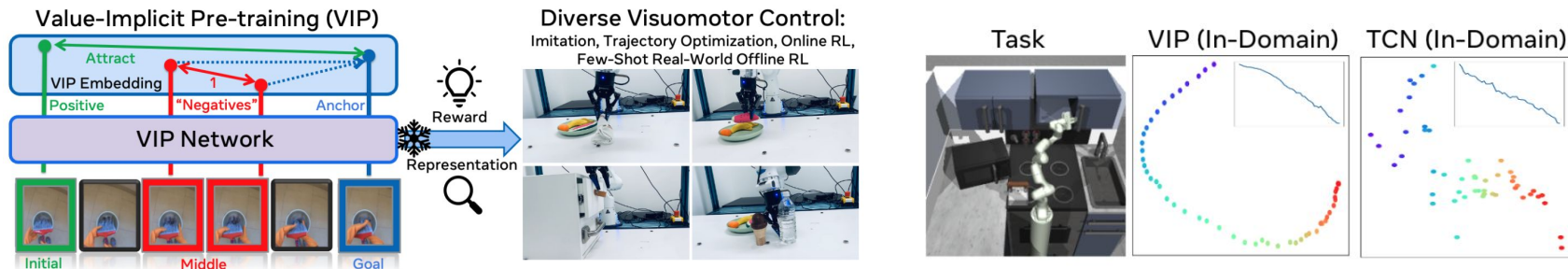
Method

VALUE-IMPLICIT PRE-TRAINING

: ANALYSIS - IMPLICIT TIME CONTRASTIVE LEARNING

- Two trends emerge:

- (1) neighboring frames are close-by in the embedding space
- (2) their distances to the last (goal) frame smoothly decrease due to the recursively defined embedding distances.



First, the explicit attraction of the initial and goal frames enables capturing long-range semantic temporal dependency as two frames that meaningfully indicate the beginning and end of a task are made close in the embedding space.

This closeness is also well-defined due to the one-step TD backup that makes every embedding distance recursively defined to be the discounted number of timesteps to the goal frame.

$$\tilde{\delta}_g(o) + \gamma V^*(\phi(o'); \phi(g)) - V^*(\phi(o), \phi(g))$$

Combined with the implicit yet structured repulsion of middle frames, this push-and-pull mechanism helps inducing a **temporally smooth and consistent representation**.

Method

VALUE-IMPLICIT PRE-TRAINING

: ALGORITHM: VALUE-IMPLICIT PRE-TRAINING (VIP)

Consider V^* as a temporal distance measure in representation space(embedding distance), it can be approximately set as a similarity metric(negative L2 distance)

$$\mathcal{L}(\phi) = \mathbb{E}_{p(g)} \left[(1 - \gamma) \mathbb{E}_{\mu_0(o;g)} [\|\phi(o) - \phi(g)\|_2] + \log \mathbb{E}_{(o,o';g) \sim D} \left[\exp \left(\|\phi(o) - \phi(g)\|_2 - \tilde{\delta}_g(o) - \gamma \|\phi(o') - \phi(g)\|_2 \right) \right] \right]$$

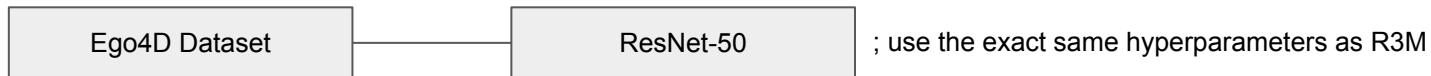
where, $V^*(\phi(o), \phi(g)) := -\|\phi(o) - \phi(g)\|_2$

Algorithm 1 Value-Implicit Pre-Training (VIP)

- 1: **Require:** Offline (human) videos $D = \{(o_1^i, \dots, o_{h_i}^i)\}_{i=1}^N$, visual architecture ϕ
 - 2: **for** number of training iterations **do**
 - 3: Sample sub-trajectories $\{o_t^i, \dots, o_k^i, o_{k+1}^i, \dots, o_T^i\}_{i=1}^B \sim D, t \in [1, h_i - 1], t \leq k < T, T \in (t, h_i], \forall i$
 - 4: $\mathcal{L}(\phi) := \frac{1-\gamma}{B} \sum_{i=1}^B [\|\phi(o_t^i) - \phi(o_T^i)\|_2] + \log \frac{1}{B} \sum_{i=1}^B \left[\exp \left(\|\phi(o_k^i) - \phi(o_T^i)\|_2 - \tilde{\delta}_{o_T^i}(o_k^i) - \gamma \|\phi(o_{k+1}^i) - \phi(o_T^i)\|_2 \right) \right]$
 - 5: Update ϕ using SGD: $\phi \leftarrow \phi - \alpha_\phi \nabla \mathcal{L}(\phi)$ * Visual encoder only
-

Experiment

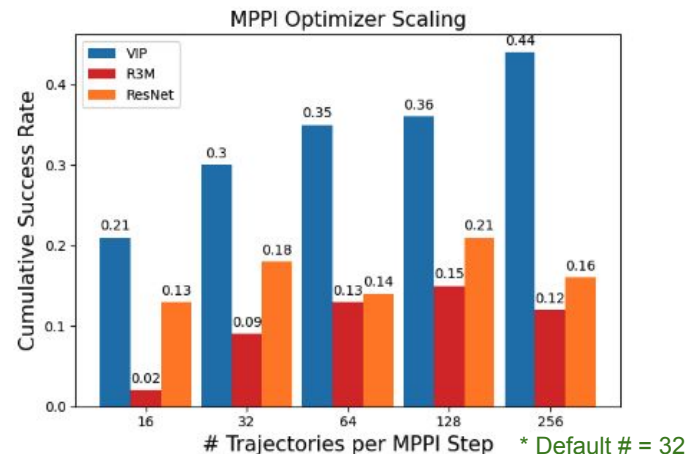
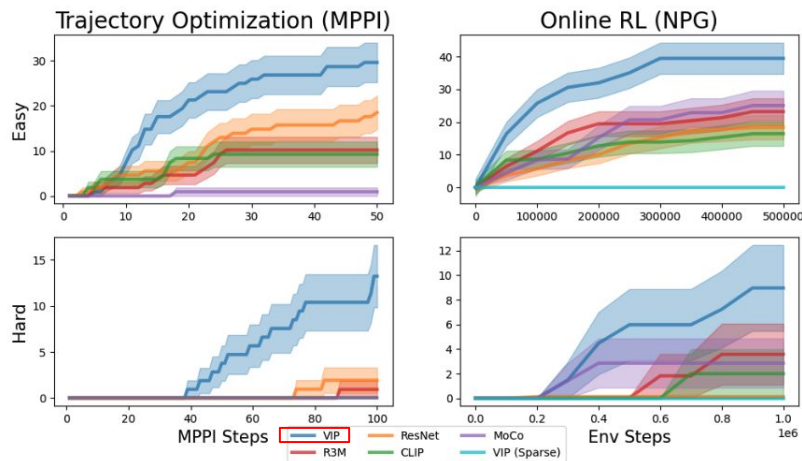
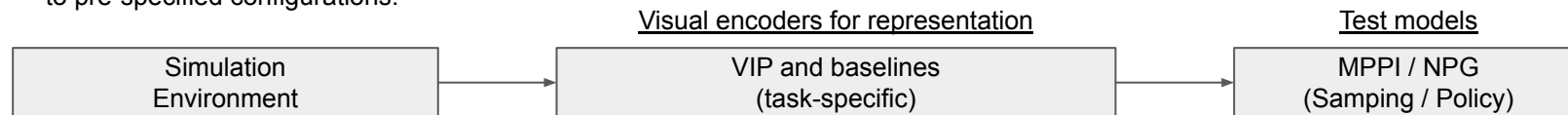
VIP Training



Trajectory optimization & Online reinforcement learning

Simulation experiments

: FrankaKitchen environment, in which a 7-DoF Franka robot is tasked with manipulating common household kitchen objects to pre-specified configurations.



Experiment

Real-World Few-shot offline reinforcement learning

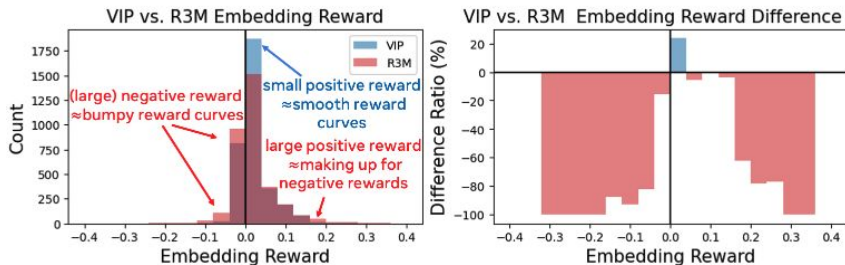
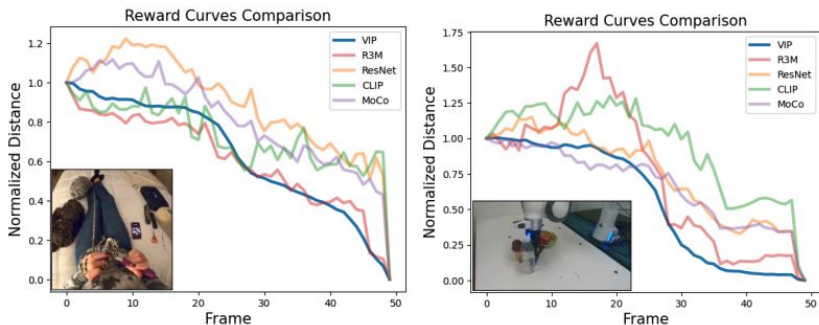
* RL simple, sample-efficient, and more effective than BC with better performance

{VIP,R3M}-{RWR,BC}

Table 1: Real-robot offline RL results (success rate % averaged over 10 rollouts with standard deviation reported).

Environment	<i>Pre-Trained</i>				<i>In-Domain</i>		
	VIP-RWR	VIP-BC	R3M-RWR	R3M-BC	Scratch-BC	VIP-RWR	VIP-BC
CloseDrawer	100 $\pm$ 0	50 $\pm$ 50	80 $\pm$ 40	10 $\pm$ 30	30 $\pm$ 46	0 $\pm$ 0	0* $\pm$ 0
PushBottle	90 $\pm$ 30	50 $\pm$ 50	70 $\pm$ 46	50 $\pm$ 50	40 $\pm$ 48	0* $\pm$ 0	0* $\pm$ 0
PlaceMelon	60 $\pm$ 48	10 $\pm$ 30	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	0* $\pm$ 0	0* $\pm$ 0
FoldTowel	90 $\pm$ 30	20 $\pm$ 40	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	0* $\pm$ 0	0* $\pm$ 0

* Robot stopped by safety check
→ overfitting



Embedding distance curves on Ego4D (L) and real-robot (R) videos.

* The bigger difference ratio
→ Bumpy learning

감사합니다.